Thermoelectric Generator – TEGs

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Introduction to Thermoelectric Generators (TEGs)

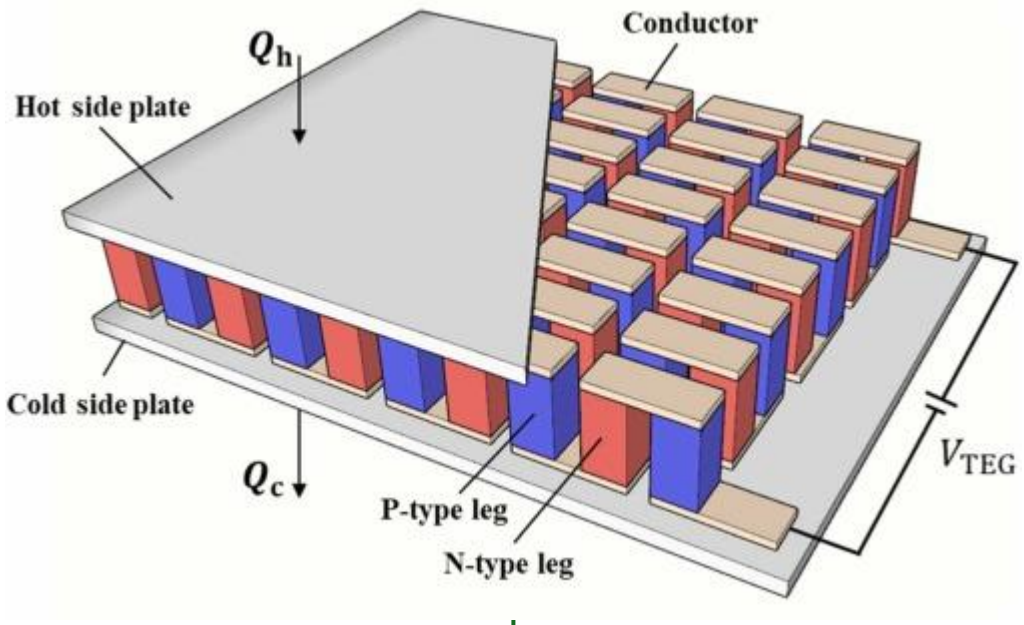
Fig. 1

What is thermoelectric Generator?

- Are devices that convert difference in temperature into electrical energy by utilizing the **Seebeck effect**.

Why are the important?

- Convert heat waste to usable energy.
- Provide sustainable and eco-friendly energy solutions.
- Operate in diverse applications from industrial to remote power generation.



- Fig. 1 Schematic overview of thermoelectric generator.

Basic Components of a TEG

1

Thermocouples (TCs):

The core of TEG systems, TCs consist of two different semiconductor materials:

- **P-type material:** This is the positive material, where holes (the absence of electrons) are the majority charge carriers.
- **N-type material:** This is the negative material, where electrons are the majority charge carriers.

2

Seebeck Coefficient:

- Each material used in the thermocouple has a property known as the Seebeck coefficient, which determines the voltage generated per unit of temperature difference between the two materials.
- **It can be calculated by:** $S = \frac{\Delta v}{\Delta T}$, where v is voltage and T tempreture.

3

Heat Source and Heat Sink:

The key to the TEG's operation is the temperature difference between the hot side and the cold side.

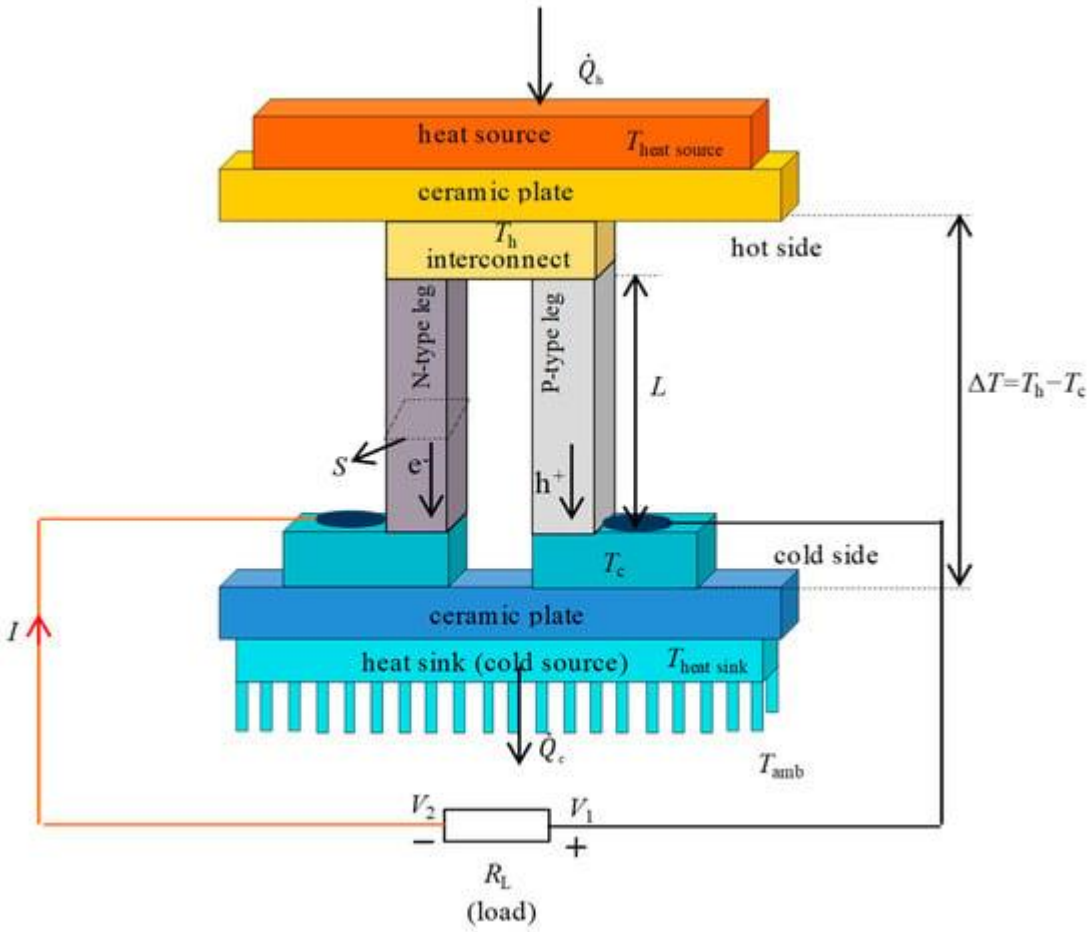
- **Heat Source**: This can be any surface or fluid at a high.
- **Heat Sink**: A heat sink or cooler is used to dissipate the heat, maintaining a temperature gradient across the TEG.

4

Electrical Circuit and Load:

The thermocouples are connected in series or parallel to form an electrical circuit. The output of the TEG is connected to a load (such as a battery or a power grid) that receives the generated electrical power.

Fig. 2



Thermoelectric Effects in Thermoelectric Generators (TEGs)

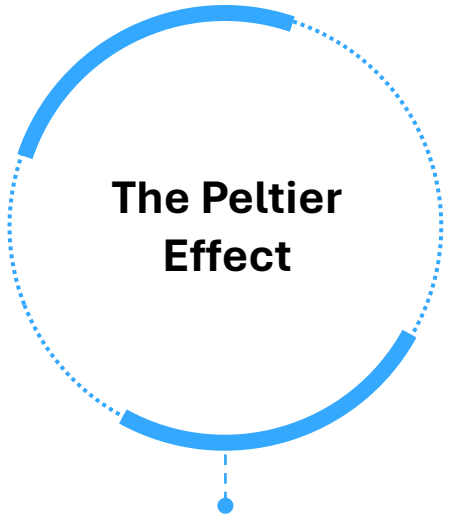


- **The Seebeck effect** is the foundation of thermoelectric generators (TEGs). It describes the generation of a voltage when there is a temperature difference across two semiconductive materials.
- **Equation:**

$$\boldsymbol{Vout = N \cdot \alpha np \cdot \Delta T}$$

Where:

- $\boldsymbol{Vout}$*: is the output voltage.*
- $\boldsymbol{\alpha np}$*: Seebeck Coefficient.*
- $\boldsymbol{\Delta T}$*: Temperature difference.*
- $\boldsymbol{N}$*: Number of thermocouples.*

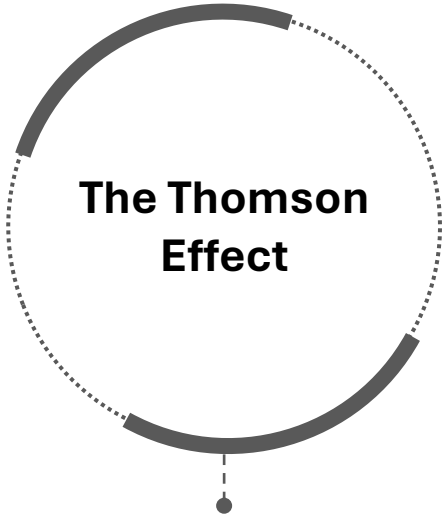


- **The Peltier effect** When current flows through a circuit made of two distinct materials, a temperature difference between the two junctions arises. The heating effect of an electric current at the junction of two different metals.
- **Equation:**

$$\boldsymbol{QPeltier = N \cdot \pi pn \cdot I}$$

Where:

- $\boldsymbol{Qpeltier}$**:** is the heat transferred.
- **N :**is the number of thermocouples.
- $\boldsymbol{\pi pn}$: re the Peltier coefficients for p-type and n-type materials.
- $\boldsymbol{I}$: is the electric current.



- **The Thomson effect** is observed when heat dissipation, absorption, or emission occurs in a conductor with a temperature gradient. The quantity of heat dissipation and absorption is governed by the conductor's temperature gradient and current density.
- **Equation:**

$$\boldsymbol{QThomson = \sigma J^2 - \mu J \frac{dT}{dx}}$$

Where:

- $\boldsymbol{\sigma}$: electrical resistivity.
- $\boldsymbol{J}$**:** current density.
- $\frac{\boldsymbol{dT}}{\boldsymbol{dx}}$: the temperature gradient.
- $\boldsymbol{\mu}$: $Thomson\ coefficient.$

Heat Transfer in Thermoelectric Generator (TEGs)

Heat transfer plays a crucial role in the operation and efficiency of TEGs. The primary modes of heat transfer in a TEG system are **conduction, convection,** and **radiation**, and each influences the temperature gradient across the thermocouples.

Coduction

- TEGs have a hot side and a cold side. Heat flows through the thermoelectric material (semiconductors) by conduction, as governed by Fourier's law. This heat transfer is critical for creating the temperature gradient that drives the **Seebeck effect**.
- The efficiency of conduction in TEGs depends on the material properties:

1. **Low k**(thermal conductivity): to maintain the temperature gradient
2. **High electrical conductivity**: (to improve power output .

$$Q\ Cond\ = -k \cdot A \cdot \frac{\Delta T}{\Delta x}$$

A: *Cross section Area.*

Δx: Thickness of the material.

*Negative sign indicates that heat flows from the hot to the cold (*decreasing temperature*).*

Convection

- On the **cold side** of a thermoelectric generator (TEG), convection is the primary mechanism for transferring heat away from the system and into the designated heat sink.
- There are to types of Convection: **Natural Convection,** heat transfer occurs naturally, **Forced Convection,** In most practical TEG systems This involves using a fan, pump.
- The rate of heat transfer via convection depends on factors such as the flow rate of the cooling fluid and its **specific heat** capacity, as well as the **surface area** of the heat sink and the **temperature difference** between the hot side and the surrounding environment.

$$Q\ Conv\ = h \cdot A \cdot (T\ surface\ - T\ fluid\ )$$

- h : Convective heat transfer coefficient ($W/m^2 \cdot K$), which depends on the fluid properties, flow type, and velocity.

Radiation

- **Radiation** is the transfer of heat in the form of electromagnetic waves.
- **Radiation** play critical role in certain applications, especially those involving high temperatures or vacuum environments.
- On the **hot side**, radiation facilitates heat transfer from the heat source to the TEG module,
- On the **cold side**, radiation can contribute to heat loss to the surrounding environment, which may help maintain the temperature gradient.

$$Q\ rad\ = \sigma \cdot \epsilon \cdot A \cdot (T1^2\ - T2^2\ )$$

σ: *Stefan-Boltzmann constant (*$5.67 \times 10^{-8} W/m^2 K^4$*).*

***ϵ:** Emissivity of the surface (dimensionless, ranges from 0 to 1).*

Working Principle of Thermoelectric Generator (TEGs)

How Do we get output Voltage?

- When a material is heated on one side, its charge carriers gain more energy, causing them to move towards the cooler side, resulting in a potential difference (voltage).

$$\boldsymbol{Vout = N \cdot \alpha np \cdot \Delta T.}$$

Where $\boldsymbol{P = IVout}$.

$$\boldsymbol{Pout = (N \cdot \alpha np \cdot \Delta T + IR)I = Vout^2 I.}$$

According to Thevenin Theorem:

$$\boldsymbol{Pmax = \frac{Vout^2}{4RTEG}}$$

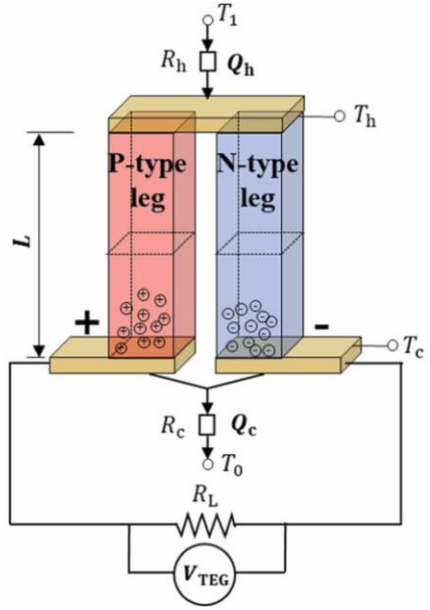
To find R TEG:

$$\boldsymbol{R\,TEG = N\left(\frac{\rho\, p\, L\, p}{\alpha\, p} + \frac{\rho\, n\, L\, n}{\alpha\, n} + 2\,\frac{\rho\, con\, L\, con}{\alpha\, con}\right)}$$

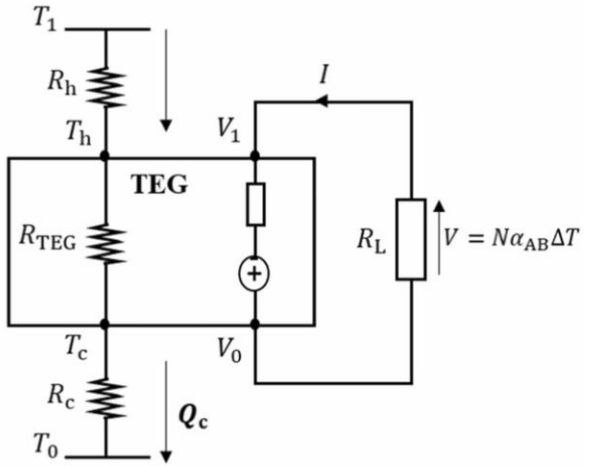
Where:

- where **ρp**, **ρn**, and **ρcon** are the **electrical resistivity** of the P- and N-type leg materials and the conductor, respectively; **Lp** and Ln are each of the **thermocouple leg lengths** crossed by the heat flow; **Lcon** is **the conductor length**, and **RL** and **RTEG** are the respective electrical resistances of the load internal TEGs.

Fig. 3



(a) Basic principle of TEG



(b) Equivalent electrical circuit

Efficiency of Thermoelectric Generator (TEGs)

Thermoelectric figure of merit

- To maximize the efficiency of TEGs, **semiconductor materials** are typically used because they offer higher Seebeck coefficients and lower electrical resistivity compared to metals.
- **Thermoelectric figure of merit**: is a dimensionless number that describes the efficiency of a thermoelectric material.
- $$ZT = \frac{\alpha np^2 \cdot \sigma \cdot T}{\kappa}$$

Where:

αnp *: Seebeck Coefficient.*

σ*: Electrical Conductivity.*

T: *Absolute Temperature (in kelvin).*

k*: Thermal Conductivity.*

TEG Efficiency:

TEGs can convert low-level heat sources into electricity. A TEG's conversion efficiency is defined by the ratio of output power to heat.

TEG Efficiency $: \eta TEG = \frac{Pout}{Qin}$

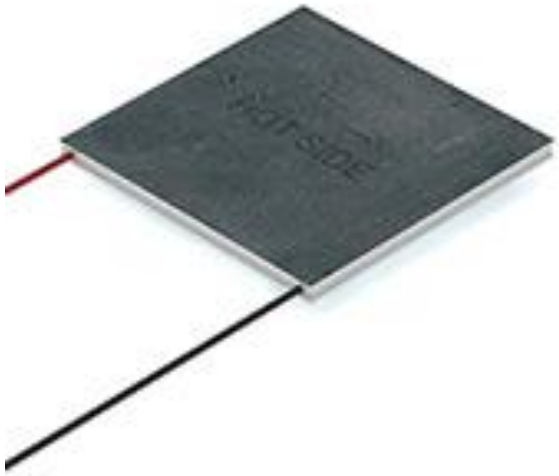
Carnot's efficiency: $\eta C = \frac{TH - Tc}{TH}$

Where:

TH: is the temperature of the hot side.

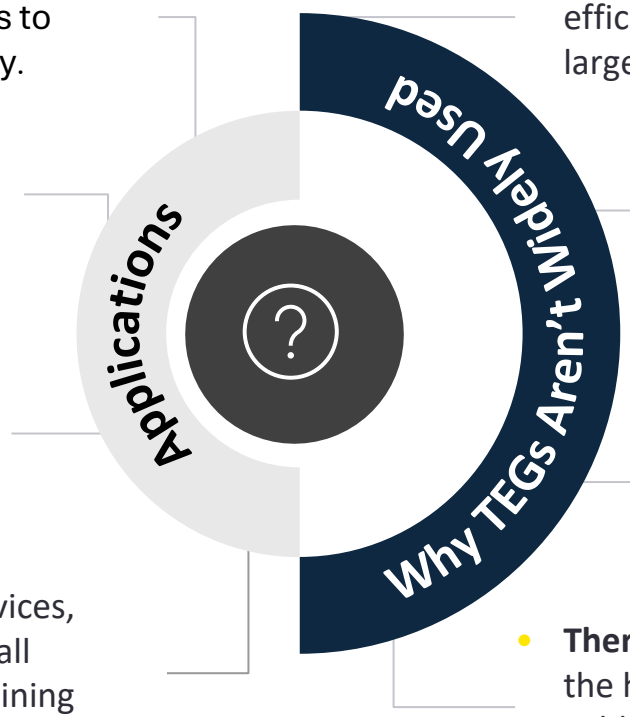
TC: is the temperature of the cold side.

Fig. 4



Thermoelectric Generator (TEGs) in Real World

- **Waste Heat Recovery:** TEGs capture waste heat from engines, power plants, and industrial processes to generate electricity, enhancing energy efficiency.
- **Space Exploration**: **TEGs power spacecraft** and **satellites**, such **as NASA's Voyager** probes, using the heat from radioactive decay.
- **Remote Power Generation**: TEGs supply electricity in off-grid locations, powering sensors, communication devices, and medical equipment..
- **Wearable Electronics**: In wearable devices, TEGs utilize body heat to generate small amounts of power, enabling self-sustaining technology.



- **Low Efficiency**: TEGs typically operate with conversion efficiencies of only **5-8%**, limiting their practicality in large-scale energy generation...
- **High Cost**: The materials used, such as bismuth telluride, are expensive and difficult to scale for mass production..
- **Limited Heat Sources**: TEGs require consistent and significant temperature differences, which aren't always readily available.
- **Thermal Management Challenges**: Managing the heat flow efficiently between the hot and cold sides is complex and affects performance.

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